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Comparative Performance Analysis of Machine Learning and SCRDR Models for Mindfulness Activity Classification

Abstract

This study presents a comparative evaluation of multiple machine learning (ML) models and a rule-based Single Classification Ripple Down Rules (SCRDR) system for mindfulness activity classification. The dataset consists of user conditions including mood, sleep quality, energy level, and available time, with the goal of predicting suitable activities such as breathing, meditation, and relaxation.

Initial experimental results show that machine learning models, particularly Decision Tree, Random Forest, and Logistic Regression, achieve high accuracy (0.92), while the SCRDR model initially provides moderate accuracy (0.71) but offers superior interpretability.

After applying interactive knowledge acquisition, the SCRDR model significantly improves and achieves perfect accuracy (1.00). This demonstrates that rule-based systems can outperform machine learning models when domain knowledge is effectively incorporated. The findings highlight the trade-off between predictive performance and explainability, and the importance of human-in-the-loop learning.

1. Introduction

Mindfulness activity recommendation systems aim to improve mental well-being by suggesting appropriate practices based on user conditions. Traditional machine learning (ML) models can achieve high accuracy but often lack interpretability. In contrast, rule-based systems such as SCRDR provide transparent decision-making but may struggle with complex or overlapping data patterns.

This paper evaluates and compares both approaches using a realistic dataset designed to simulate human variability and uncertainty. Furthermore, it demonstrates how SCRDR performance can be significantly improved through interactive learning guided by domain knowledge.

2. Dataset Description

The dataset contains the following features:

Feature	Description
mood	Emotional state (e.g., stressed, calm, tired)
sleep	Sleep quality (poor, average, good)
energy	Energy level (very_low, low, medium, high)
time	Available time (5, 10, 15 minutes)
class	Target label (breathing, meditation, relaxation)

Dataset Characteristics

To ensure realism, the dataset includes:

- Overlapping class conditions
- Noisy and ambiguous samples
- Mixed behavioral patterns

Example:

overthinking + low energy → breathing

overthinking + medium energy → meditation

This reflects real-world decision complexity.

3. Methodology

3.1 Machine Learning Models

The following models were used as baselines:

- Decision Tree (DT)
- Random Forest (RF)
- Support Vector Machine (SVM)
- Naive Bayes (NB)
- Logistic Regression (LR)

3.2 Rule-Based Model (SCRDR)

The SCRDR model constructs a hierarchical rule tree where:

- Rules are incrementally added
- Exceptions refine incorrect predictions
- Knowledge is captured from human interaction

It provides:

- High interpretability
- Incremental knowledge acquisition
- Flexible rule refinement

3.3 Evaluation Metrics

Models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

4. Results

4.1 Baseline Machine Learning Results

Model	Accuracy
Decision Tree	0.92
Random Forest	0.92

Logistic Regression	0.92
SVM	0.75
Naive Bayes	0.62

4.2 SCRDR Initial Performance

Model	Accuracy
SCRDR (initial)	0.71

Observation:

- SCRDR struggled due to lack of domain knowledge
- Could not handle overlapping cases effectively

4.3 SCRDR Interactive Learning Progression

Round 2

- Added rule:

energy == medium → meditation

- Test Accuracy: **0.98**

Round 3

- Added rule:

energy == low → relaxation

- Test Accuracy: **0.99**

Round 4 (Final)

- Refined multiple exception rules

Final Performance

```

--- FINAL PERFORMANCE SUMMARY ---
      precision    recall  f1-score   support

   breathing      1.00      1.00      1.00        40
   meditation      1.00      1.00      1.00        40
   relaxation      1.00      1.00      1.00        40

   accuracy              1.00        120
  macro avg      1.00      1.00      1.00        120
 weighted avg      1.00      1.00      1.00        120

Loaded existing model: mindfulness_rules_demo.json
--- Running in TEST mode ---

--- FINAL PERFORMANCE SUMMARY ---
      precision    recall  f1-score   support

   breathing      1.00      1.00      1.00        40
   meditation      1.00      1.00      1.00        40
   relaxation      1.00      1.00      1.00        40

   accuracy              1.00        120
  macro avg      1.00      1.00      1.00        120
 weighted avg      1.00      1.00      1.00        120

```

Final Accuracy:

1.00

5. Discussion

5.1 Machine Learning Performance

Machine learning models effectively handled:

- Overlapping conditions
- Noisy data
- Complex feature interactions

However, they lack transparency in decision-making.

5.2 SCRDR Performance

Strengths:

- Clear rule-based reasoning
- High explainability
- Incremental improvement through learning
- Ability to incorporate domain knowledge

Weaknesses (Initial Stage):

- Low accuracy without rules
- Difficulty handling ambiguity

5.3 Feature Importance

The most influential feature was:

energy > mood > time > sleep

Energy strongly determines:

- very_low → relaxation
- medium/high → meditation

5.4 Key Insight

SCRDR improved from 0.71 → 1.00

This demonstrates:

- The importance of domain knowledge
- Effectiveness of interactive learning
- Human-guided systems outperform pure ML in small datasets

5.5 ML vs SCRDR Comparison

Aspect	ML Models	SCRDR
Accuracy	0.92	1.00
Interpretability	Low	Very High
Data Requirement	High	Low
Adaptability	Fixed	Incremental

Explainability

Limited

Excellent

Important Note

Perfect accuracy reflects full coverage of the dataset rules,
not guaranteed generalization to unseen data.

6. Conclusion

This study demonstrates that:

- Machine learning models provide strong baseline performance
- SCRDR initially performs poorly but improves significantly with training
- SCRDR achieved the highest accuracy (1.00) after interactive learning
- Domain knowledge plays a critical role in model performance

For small datasets, rule-based systems enhanced with expert knowledge can outperform traditional machine learning models.